

Real Impact-Factor Analysis

of Academic Journals — refining the IF with graph-learning



PROJECT CODE

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STUDENTS

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BACKGROUND & MOTIVATION

The Journal **Impact Factor (IF)** — the average number of citations a journal's recent articles receive — governs prestige, hiring, promotion, tenure and funding, yet it is computed from **citation counts taken exactly as collected**.

Because a single number carries so much weight, citation networks have become a target for deliberate inflation:

- **Citation cartels** — groups of journals or authors that reciprocally cite one another to lift their scores
- **Excessive self-citation** — a journal padding its own count by citing itself
- **Coercive citation** — editors or reviewers pressuring authors to add unnecessary references

THE CHALLENGE

The conventional IF treats every citation as equally valid — it never asks whether an individual link reflects *genuine* scientific influence.

Global anomaly detectors scan the network as a whole and routinely miss subtle, locally-targeted manipulation. Independent studies estimate that more than **two-thirds** of the references in a typical paper are **non-essential** to its argument.

Our goal: a citation metric that rewards authentic influence while neutralising noise, padding and coordinated manipulation.

HYPOTHESIS

Genuine citations are structurally robust; manipulated citations are fragile

A legitimate citation is reinforced by the surrounding network — shared references, co-citations and indirect paths — so it can be **rebuilt from the graph alone**. A manufactured citation has no such support, so once removed it **cannot be reconstructed**. That difference is what we exploit.

THE SOLUTION — BUILDING A RELIABLE NETWORK

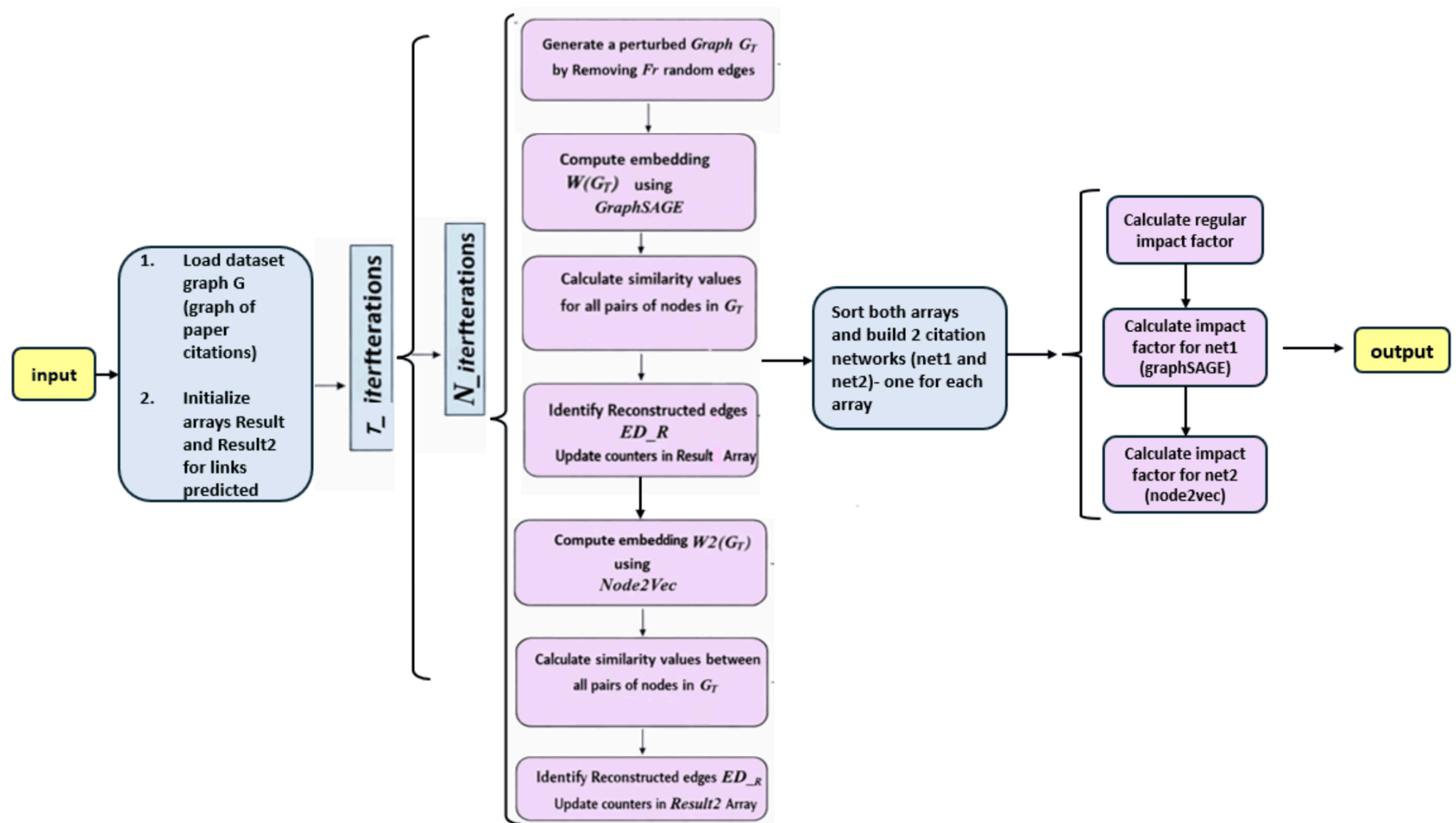


Figure 1 — End-to-end pipeline. The graph is perturbed, re-embedded with both models and reconstructed over N iterations and each 2-year window before the two impact factors are computed.

REGULAR IMPACT FACTOR

$$IF_Y = \frac{\text{Citations in } Y \text{ to items published in } Y-1 \text{ and } Y-2}{\text{Citable items published in } Y-1 \text{ and } Y-2}$$

Counts **every reported citation**, regardless of whether it is legitimate.

IMPROVED IMPACT FACTOR

$$IF_Y^\star = \frac{\text{Reliable citations in } Y \text{ to items published in } Y-1 \text{ and } Y-2}{\text{Citable items published in } Y-1 \text{ and } Y-2}$$

Numerator counts **only the reliable citations** — links that survive reconstruction — not every citation on record.

THE MODELS — HOW EDGES ARE REBUILT

GraphSAGE

Learns a vector for every paper by **sampling and aggregating the features of its neighbours** over successive hops, so a node is described by the company it keeps. Because it learns an aggregation rule rather than memorising the graph, it **generalises to new and evolving nodes** without retraining — well suited to a citation network that grows each year.

Node2Vec

Generates biased **random walks** through the graph and feeds them, like sentences, to a skip-gram model that places structurally similar papers near one another. Two parameters, p and q , tune the walk between **breadth-first** exploration (local community) and **depth-first** exploration (broader structural role), capturing both kinds of similarity.

Link Prediction

Given two learned embeddings, the **cosine similarity** between them scores how likely an edge is; when it exceeds the threshold T_r , the edge is declared reconstructed. Three controls govern the procedure: iterations N_{iter} , the removed fraction F_r , and the reliability threshold T_r — the parameter that ultimately decides which links survive.

RESULTS & VALIDATION

1 The Improved IF sits consistently below the conventional IF

Across **every (journal, year) pair**, the Improved IF fell below the Regular IF — exactly what is expected once unsupported links are stripped out. **GraphSAGE** filtered most aggressively (Improved IF $\approx$ one-quarter to one-fifth of the conventional value); **Node2Vec** was more conservative ($\approx$ one-half).

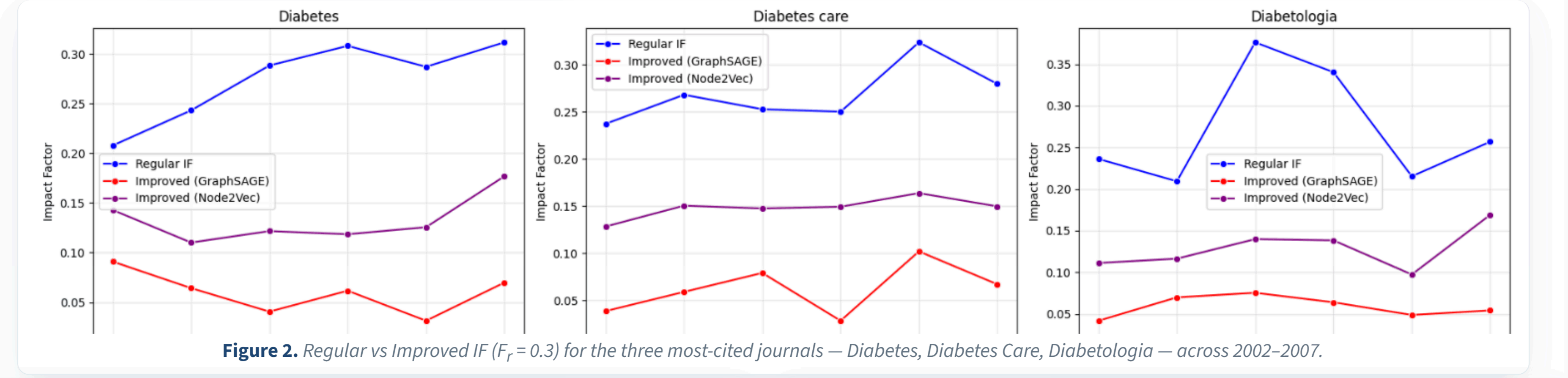


Figure 2. Regular vs Improved IF ($F_r = 0.3$) for the three most-cited journals — Diabetes, Diabetes Care, Diabetologia — across 2002–2007.

2 The filter omits randomly-generated (fabricated) citations

A control injected **10% synthetic edges** each year. Almost none cleared the cutoff, while authentic edges did — the filter is **selective, not arbitrary**.

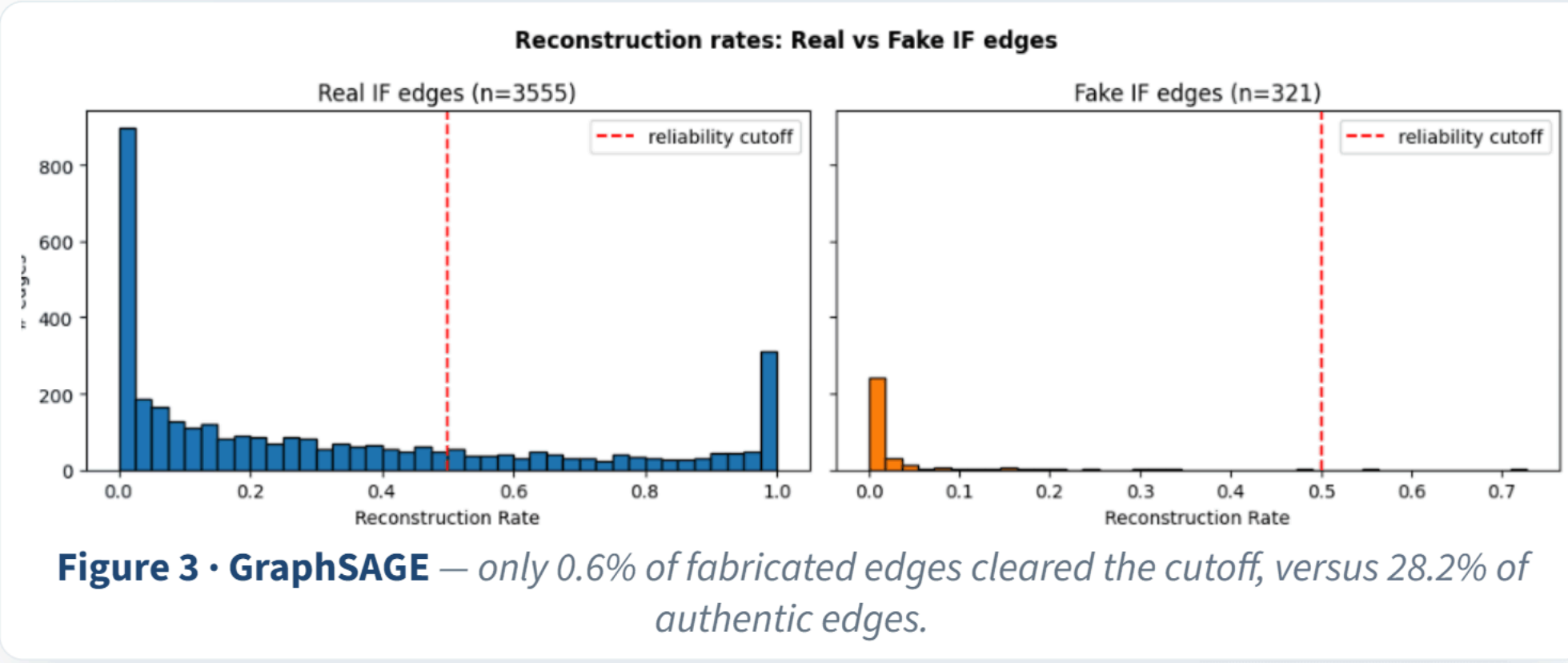


Figure 3 • GraphSAGE — only 0.6% of fabricated edges cleared the cutoff, versus 28.2% of authentic edges.

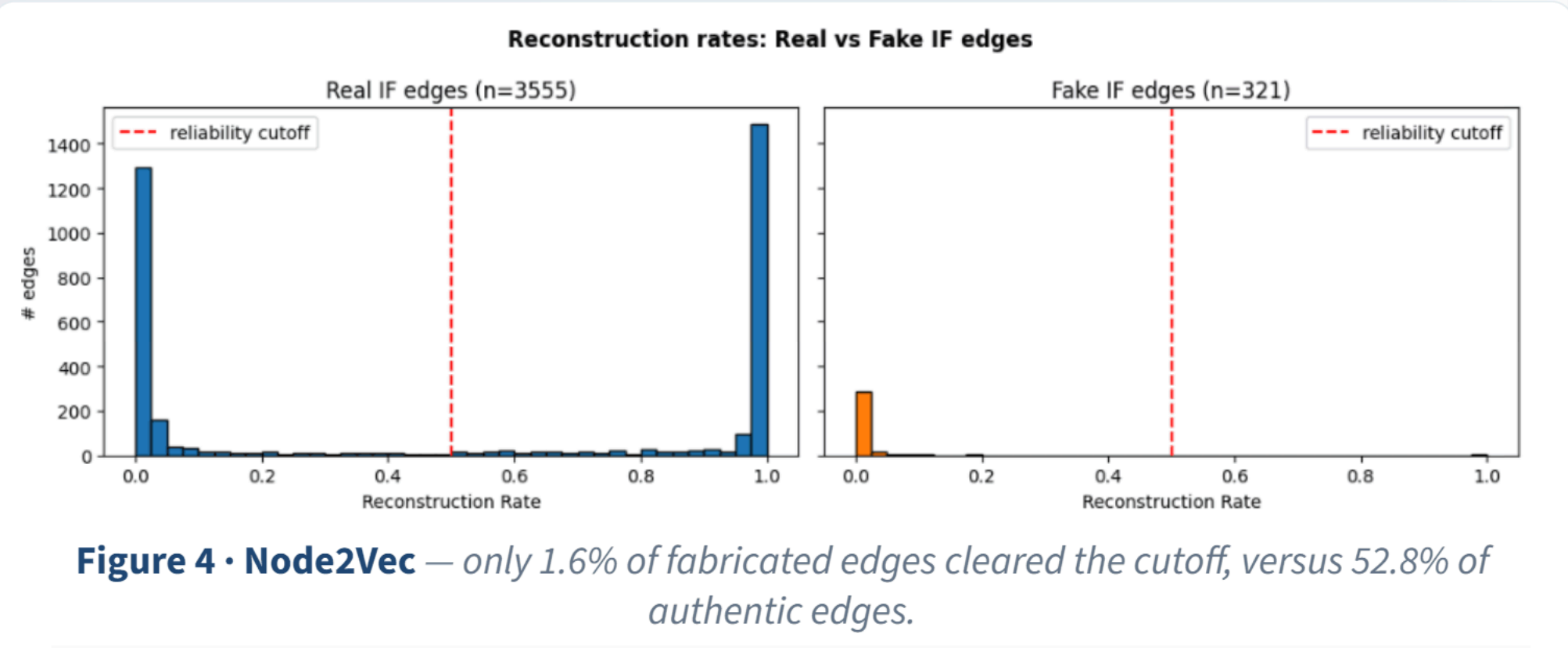


Figure 4 • Node2Vec — only 1.6% of fabricated edges cleared the cutoff, versus 52.8% of authentic edges.

3 Insensitive to how much is removed; governed by the threshold

Varying the removed fraction F_r from 30% to 70% barely changes the result. The decisive lever is the reliability threshold T_r , calibrated per embedding.

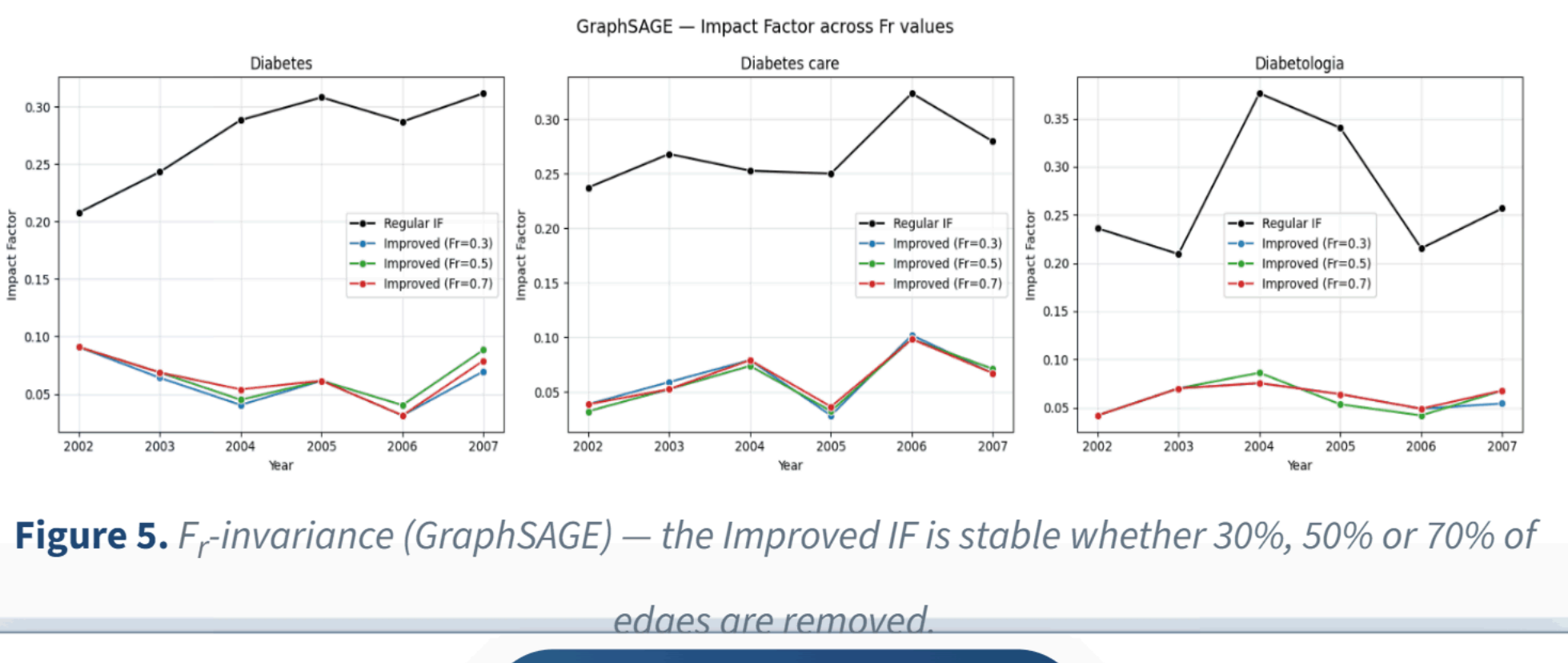


Figure 5. F_r -invariance (GraphSAGE) — the Improved IF is stable whether 30%, 50% or 70% of edges are removed.

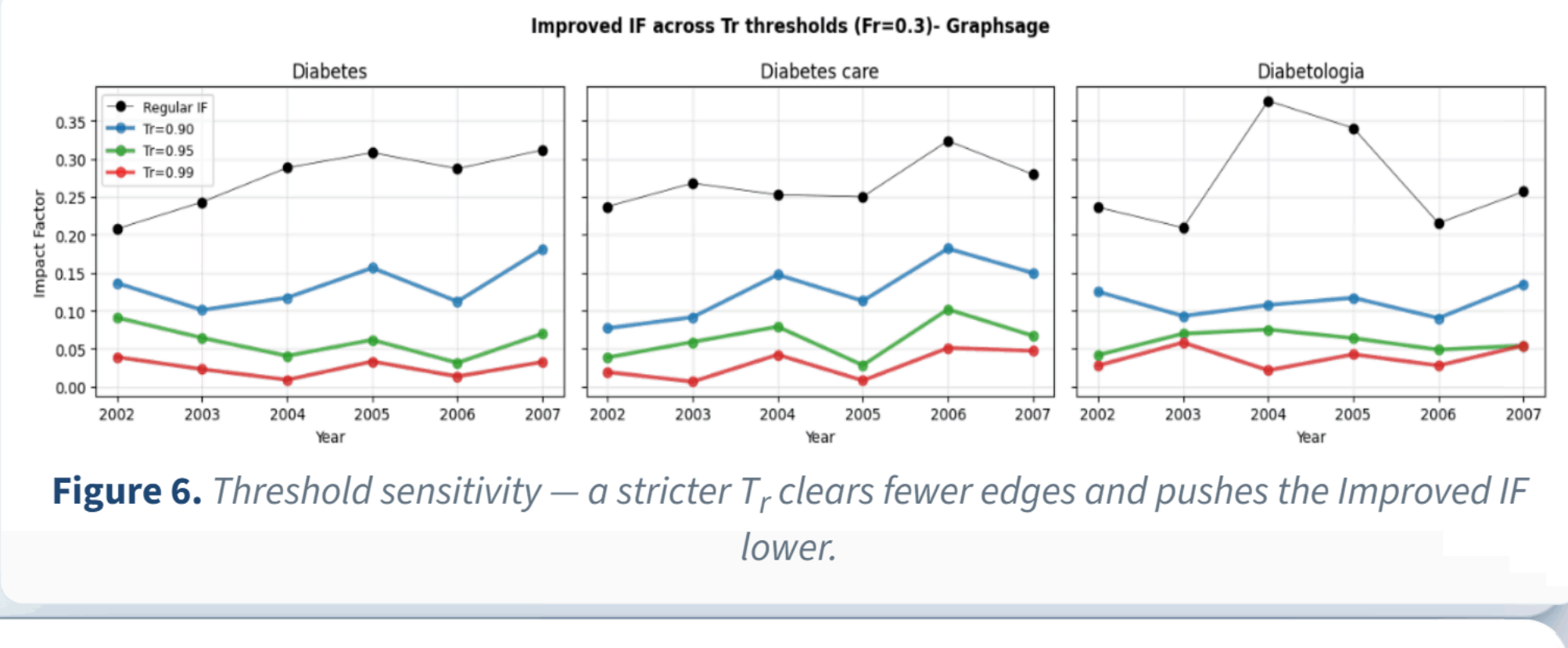


Figure 6. Threshold sensitivity — a stricter T_r clears fewer edges and pushes the Improved IF lower.

CONCLUSIONS

- On the PubMed diabetes network, both embeddings independently push the Improved IF below the Regular IF in every configuration tested.
- The fabricated-edge control confirms the filtering is **genuine and targeted**, not indiscriminate thinning.
- A substantial share of reported citations is **not robustly supported** by the surrounding network structure.
- Outcomes are invariant to F_r but governed by T_r , calibrated per embedding — a clear, tunable reliability dial.